A Global-to-Local Evolutionary Algorithm for Hyperspectral Endmember Extraction

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***Abstract*— Recently, evolutionary algorithms (EAs) have shown their promising performance in solving the hyperspectral endmember extraction (EE) task. Despite that, most of the existing EA-based EE algorithms mainly take advantage of the global search capability of evolutionary computation. A few of them focus on the hyperspectral EE task itself, which is a sparse large-scale problem with constraint. To fill the gap, in this article, a global-to-local EA (GL-EA) is proposed, where the global and local search is performed sequentially to extract the endmembers effectively. Specifically, in the first global search stage, two complementary solution generation strategies, including asymmetric flip-based solution generation and spectral angle distance (SAD)-based solution repair, are designed, with which the sparse large-scale search space of hyperspectral EE is fully explored and the endmembers that satisfy the constraint could be achieved. Then, in the second stage, a perturbation-based local search is suggested, which further enhances the quality of the obtained endmembers. In addition, an endmember repetition-based solution selection strategy is also developed for both global and local search stages, by using which good solutions can be selected effectively during the evolution. Experimental results on different hyperspectral datasets demonstrate that when compared with the state-of-the-art EE algorithms, the proposed GL-EA could extract the endmembers with higher quality.**

***Index Terms*— Endmember extraction (EE), evolutionary algorithm (EA), global-to-local search, hyperspectral image (HSI), multiobjective optimization.**

NOMENCLATURE

NP	Size of population.
M	Number of endmembers.
N	Number of pixels.
L	Number of spectral bands.
ps_1	Good parent solution.
ps_2	Bad parent solution.
os	Offspring solution.
s	Solution.
p^{0-1}	Probability of flipping from 0 to 1.
p^{1-0}	Probability of flipping from 1 to 0.
$\{e_m\}_{m=1}^{M}$	Endmember spectra set in ps_1.
$\{e_j\}_{j=1}^{M}$	Endmember spectra set in ps_2.
$\{\text{pos}_i\}_{i=1}^{M}$	Endmember position set in ps_1.
M_i	ith endmember in a solution.
r	Parameter for determining the range of perturbation.
K	Parameter for dividing the global search and the local search.
maxgen	Maximum generation.
$\mathbf{P}_t$	tth parent population.
$\mathbf{O}_t$	tth offspring population.

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I. INTRODUCTION

HYPERSPECTRAL remote sensing images contain a large number of spectral bands, which contain rich features reflecting the components in the scene, enabling hyperspectral images (HSIs) to be advantageous in object detection over multispectral remote sensing images [1], [2], [3]. However, due to the limited spatial resolution of imaging spectrometers, some pixels may contain more than one type of ground object signatures in HSIs, and these pixels are termed mixed pixels [4], [5], [6]. Mixed pixels bring many difficulties to object detection on HSIs [7]. Hyperspectral unmixing (HU) can effectively solve the problem of mixed pixels by searching the basic ground objects that make up the mixed pixels and calculating the proportion of each basic ground object in mixed pixels. In other words, HU aims to obtain the pure pixel set (endmembers, also known as pixels that only contain one type of ground object) and the corresponding fraction set (abundance) [8], [9], [10]. As a key component of HU, endmember extraction (EE) is used to find out the pure pixels, which has a great influence on the whole performance of HU [11], [12], [13], [14].

To tackle the EE problem, many algorithms have been suggested. According to the adopted techniques, these algorithms can be roughly classified into four categories [15].

1) *Pure Pixel Assumption-Based EE Algorithms:* In this category, the assumption that HSIs have pure pixels for each ground object is made. For example, Boardman et al. [16] proposed a pixel purity index (PPI)

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approach, where data projection was used to find the positions of endmembers and could achieve good EE results. Chang and Plaza [17] proposed a fast iterative PPI (FIPPI) algorithm, which constructed an appropriate initial set of endmembers and adopted virtual dimensionality to efficiently estimate the endmembers number. Other work that belongs to this category can be found in [18], [19], and [20].

2) *Minimum Volume Model-Based EE Algorithms:* In this category, the pure pixel assumption was not adopted. Instead, a simplex is designed in the feature space so that all pixels in the image can be located inside the simplex. Thus, the optimization task is to find such a simplex with the minimal volume. Fuhrmann [21] proposed a simplex shrink-wrap algorithm (SSWA), which searched the simplex through shrinking the volume with gradient descent. Li and Dias [22] proposed a minimum volume simplex analysis (MVSA) algorithm, which fit a minimum volume simplex to the hyperspectral data and constrained the abundance fractions to belong to the probability simplex. Besides, more work could also be found in [23] and [24].
3) *Spatial Information-Based EE Algorithms:* The spatial correlation of pixels in HSIs is strong. Based on this fact, Rogge et al. [25] proposed a spatial–spectral EE (SSEE) algorithm, where the spectral contrast of low contrast endmembers was increased by analyzing a scene in parts. Zortea and Plaza [26] further proposed a spatial preprocessing (SPP) algorithm. In their proposed algorithm, for each pixel, a scalar indicating the spectral similarity with neighboring pixels was defined to measure the weight of the spectral information, by using which the endmembers with good quality could be obtained. In addition, the work in [27], [28], and [29] also belong to this category.
4) *Statistical Model-Based EE Algorithms:* The work in [30], [31], and [32] assumed that the spectral components had statistically independent and constructed unsupervised methods based on independent component analysis (ICA) to solve the EE problem. Moreover, Nascimento and Dias [33] proposed a dependent component analysis (DECA) algorithm, which was a statistical and fully automatic EE algorithm based on linear spectral mixing model (LSMM) [6] and was suitable for highly mixed HSIs.

Despite that the above EE algorithms can obtain the endmembers in HSIs, most of them are based on geometrical or statistical hypotheses. When there are noises or outliers in HSIs, these methods are prone to get trapped into local optimum and cannot effectively extract accurate endmembers [34]. To tackle the issue, recently, a number of evolutionary computation-based algorithms (EAs) have been developed to solve the EE problem. For instance, Zhang et al. [35] first regarded the EE problem as an intelligent optimization problem and proposed a discrete particle swarm optimization (DPSO) algorithm, which could acquire higher extraction accuracy than traditional EE algorithms. Inspired by the superiority of DPSO in solving the EE problem, more EA-based algorithms have been proposed [36], [37], [38], [39], [11], [40], [41], [42], and gradually become a research hotspot in the EE field.

Although these EAs have shown promising performance in obtaining good endmembers, these algorithms utilize the global search ability of EA and mainly focus on using different EA frameworks (such as particle swarm optimization (PSO) [11], [35], [42], ant colony optimization (ACO) [37], and differential evolution (DE) [36]) to design EE algorithms, and a few of them take a closer look at the hyperspectral EE problem itself. For example, the pixels' number in an HSI is often very large, usually more than 10 000. On the contrary, the endmembers number is fixed and much smaller. Thus, when adopting evolutionary optimization, the hyperspectral EE problem is essentially a sparse large-scale optimization problem with the constraint of fixed endmembers' number. These characteristics of the EE problem bring great challenges to the existing EA-based EE algorithms. To fill the gap, in this article, a global-to-local evolutionary algorithm (GL-EA) is proposed, and the main contributions of this work are summarized as follows.

1) The hyperspectral EE problem is first viewed as a sparse large-scale and constrained optimization problem. To tackle this problem, a global-to-local search idea is proposed to obtain the optimal endmembers in a gradual way. The suggested idea includes two stages: global search and local search. Global search aims to fully explore the sparse large-scale search space and satisfy the fixed endmembers' number constraint. Local search is used to refine the global search results and further enhance the qualities of the final solutions.
2) To implement the suggested global-to-local search idea, an evolutionary algorithm (EA), named GL-EA, is proposed. Specifically, two complementary solution generation (asymmetric flip-based solution generation and spectral angle distance (SAD)-based solution repair) strategies are designed in the global search stage. Meanwhile, a perturbation-based local search is developed in the second stage. In addition, an endmember repetition-based solution selection strategy is also suggested in GL-EA, with which good solutions could be picked out during the evolution.
3) The effectiveness of the proposed GL-EA is validated on three widely used HSI sets. Experimental results show that in comparison with the representative non-EA- and EA-based algorithms, the GL-EA can extract the endmembers with higher qualities.

The rest of this article is organized as follows. The related work is reviewed in Section II. Then, the details of the proposed GL-EA are presented in Section III. The experimental results and analysis of the proposed GL-EA are reported in Section IV. Finally, we conclude this article and discuss the future work in Section V.

II. Related Work

In this section, we will review the related work of GL-EA. Specifically, first, we will present the LSMM for the hyperspectral EE problem. Then, the preliminary work of

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multiobjective optimization is introduced since the proposed GL-EA is a multiobjective evolutionary algorithm (MOEA). Finally, we focus on the state-of-the-art MOEAs for the hyperspectral EE problem, which are closely related to our work.

A. Linear Spectral Mixing Model

Due to its simplicity and effectiveness, the LSMM [6] is commonly utilized to deal with the EE problem [39], [41]. LSMM assumes that multiple scattering among endmembers is negligible, and the spectrum of each pixel can be expressed as a linear mixture of endmembers spectra and weighted by the fractional area coverage (abundance) of each endmember within the pixel. The LSMM model can be described as follows:

$$\boldsymbol{R}_{L\times N} = \boldsymbol{E}_{L\times M}\boldsymbol{A}_{M\times N} + \boldsymbol{\varepsilon}_{L\times N} \tag{1}$$

where L, N, and M indicate the number of bands, pixels, and endmembers, respectively, $\boldsymbol{R}$ represents the image matrix, $\boldsymbol{E} = [e_1, e_2, \ldots, e_M]$ denotes the endmember matrix, $\boldsymbol{A} = [a_1, a_2, \ldots, a_N]$ represents the abundance matrix, and ε represents the error matrix. Once the endmembers are determined, the corresponding abundance of each endmember can be calculated. In this process, two constraints should be considered, including the abundance sum-to-one constraint (ASC) and the abundance nonnegative constraint (ANC). Similar to many EE work [38], [39], [40], [41], in this article, the nonnegative constrained least squares (Ncls) method [43] is used to calculate abundances.

B. Multiobjective Optimization

Multiobjective optimization aims to simultaneously optimize two or more objectives and can be described as

$$\min F(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_n(\mathbf{x}))^T, \quad \text{s.t. } \mathbf{x} \in \Omega \tag{2}$$

where $\mathbf{x}$ represents the solution (decision vector) and Ω denotes its feasible space. $f_i(\mathbf{x})$ represents the ith objective to be optimized. For most multiobjective optimization problems (MOPs), the various objectives are often conflicting with each other, so it is difficult to make all the objectives reach the optimum at the same time. The multiobjective optimization aims to obtain a set of compromised solutions, which are called the Pareto optimal set. Just take the minimization problem as an example, a solution $\mathbf{x_u}$ dominates another solution $\mathbf{x_v}$ if and only if

$$\forall i = 1, 2, \ldots, n \quad f_i(\mathbf{x}_\mathrm{u}) \le f_i(\mathbf{x}_\mathrm{v}). \tag{3}$$

A solution $\mathbf{x}^*$ in Ω is called a Pareto optimal solution if and only if $\mathbf{x}^*$ is not dominated by other solutions. All the Pareto optimal points constitute a set called Pareto optimal set [44], and their corresponding projections in the objective space are called the Pareto front (PF). In addition, hypervolume (HV) [45], [46] is commonly used to evaluate PFs of different optimization methods.

In the last two decades, EA has shown to be a rich framework in solving MOPs, and a lot of promising MOEAs have been developed, including NSGA-II [47], MOEA/D [48], and IBEA [49]. Meanwhile, MOEAs have also been applied to solve MOPs in different applications, such as community detection [50], [51], data mining [52], [53], feature selection [54], [55], and vehicle routing problem [56], [57]. Recently, some researchers proposed to design MOEAs to solve the EE problem. In Section II-C, we will review these algorithms in detail.

C. MOEAs for EE

As mentioned before, there are many EA-based methods that have shown good performance in solving EE problems. These EA-based algorithms include single-objective EA-based methods and MOEA-based methods. For single-objective EA-based methods, the work [11], [42] aimed to optimize the volume inverse, while the work [35], [36] focused on optimizing the objective of the root-mean-square error (RMSE). However, in the work above, only one objective (the volume inverse or the RMSE) was considered. Compared with the single-objective EA-based methods, the MOEA-based methods have at least two merits: 1) by utilizing the multiobjective optimization technique, multiple objectives (e.g., the volume inverse and RMSE) in EE can be integrated without using any tradeoff parameter and 2) MOEA-based EE algorithms can provide multiple solutions in one run and satisfy different needs of real applications. Therefore, recently, the MOEA-based EE algorithms have been paid much more attention to.

Liu et al. [38] proposed a multiobjective DPSO (MODPSO) algorithm to tackle the EE problem, which was the first time to introduce evolutionary multiobjective optimization into the EE problem. Their two optimization objectives were the volume inverse and the RMSE. To solve the binary objective optimization problem, MODPSO was designed under the framework of PSO. Experimental results on two real HSIs showed the superiority of the proposed MODPSO over several single-objective EE methods (N-FINDR, VCA, DPSO, and so on).

To further enhance the performance, Tong et al. [39] proposed an improved MODPSO (IMODPSO) for the hyperspectral EE problem. In IMODPSO, the volume inverse and the RMSE were still taken as the optimization objectives. Moreover, a restart operator was designed to remove the duplicate particles and increase the particles' search ability. For the discarded solutions, an adaptive archiving mechanism was also proposed to increase the population's diversity. Experimental results showed that IMODPSO not only had better performance than the traditional non-EA algorithms but also outperformed some representative EA-based methods.

Different from MODPSO and IMODPSO which take the PSO as the basic framework, Cheng et al. [40] suggested an adaptive reference point-based nondominated sorting genetic algorithm, named ANSGA-III, which was under the framework of NSGA-III. In ANSGA-III, to tailor for the EE task, reference points were adaptively updated to guide the evolution. Thus, the population's diversity could be maintained to a certain extent. Experimental results on three HSIs

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indicated that ANSGA-III had more powerful search ability than many EA-based methods. Meanwhile, ANSGA-III was also robust to outliers.

Moreover, Tong et al. [41] suggested an MOEA [$(\mu + \lambda)$MODE] for the EE problem, which was based on the DE algorithm. In the $(\mu + \lambda)$MODE algorithm, ranking multiple mutations was adopted to create the mutant vectors via the scaling factor pool. In addition, a fast nondominated sorting approach and a $(\mu + \lambda)$ selection strategy were used to enhance the quality of the extracted endmembers. Experimental results on three real HSIs revealed that $(\mu + \lambda)$MODE had better performance than the existing state of the arts.

In summary, from the abovementioned methods, it can be noticed that MOEAs have shown better performance in solving EE problems in comparison with the traditional EE algorithms and single-objective EAs. However, the existing MOEA-based EE algorithms mostly focus on choosing different evolutionary optimization frameworks (e.g., PSO, NSGA-III, and DE) and designing the corresponding global search algorithms. Actually, the EE problem is a sparse large-scale and constrained optimization problem. In the aforementioned work, this characteristic of the EE problem has not been considered. To tailor for the sparse large-scale and constrained search space of the EE problem, a GL-EA is proposed to search the optimal solutions in a gradual way. In Section III, we will discuss our algorithm in detail.

III. Global-to-Local Evolutionary Algorithm

Before we present the details of the proposed GL-EA, first, we list the main notations used in this article in the Nomenclature, which could help the readers better understand our GL-EA.

The proposed GL-EA aims to minimize two classic metrics simultaneously: the volume inverse $[f_1(X)]$ and the RMSE $[f_2(X)]$, which are widely used in EE tasks [40], [41] and formally defined as

$$\begin{cases} f_1(\mathrm{X}) = \dfrac{1}{\mathrm{volume}(\boldsymbol{A}(\mathrm{X}))} = \dfrac{(M-1)!}{\left|\det\begin{bmatrix} 1 & 1 & \dots & 1 \\ \boldsymbol{a}_1 & \boldsymbol{a}_2 & \dots & \boldsymbol{a}_M \end{bmatrix}\right|} \\ f_2(\mathrm{X}) = \mathrm{RMSE}(\boldsymbol{r}, \hat{\boldsymbol{r}}) = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N}\sqrt{\dfrac{1}{L}\left\|\boldsymbol{r}_i - \hat{\boldsymbol{r}}_i\right\|_2^2}. \end{cases} \tag{4}$$

In formula (4), X represents a solution (an individual) in the population that records the endmembers position information. $\boldsymbol{A}(X)$ is the endmembers matrix. M represents the number of endmembers, and $\{\boldsymbol{a}_i\}_{i=1}^{M}$ denotes the endmembers data after minimum noise fraction (MNF) dimension reduction, while $\boldsymbol{r}$ and $\hat{\boldsymbol{r}}$ represent the original HSI pixel and the remixed image pixel obtained from the solution X, respectively. N denotes the total number of pixels, and L is the number of spectral bands.

Like most EA-based hyperspectral EE algorithms, the solution X in the proposed method is also represented with the binary encoding scheme, which can be represented as

$$X = \left\{ (x_1, x_2, \dots, x_N) \mid x_i \in \{0, 1\}, \sum_{i=1}^{N} x_i = M \right\} \tag{5}$$

where x_i corresponds to the ith pixel of the original HSI in the horizontal raster-scanning order. If x_i equals 1, it means that the ith pixel of the original HSI is selected as the endmember, and otherwise, it means not selected.

By taking a close look at formulas (4) and (5), we can find that the hyperspectral EE problem is actually a sparse large-scale optimization problem, as the length of the solution equals the pixels number (N) of the HSI, which is much larger than the number of endmembers (M). Just taking the famous Urban dataset as an example, the solution's length (N) is 94,249, and however, its endmembers number (M) is only 6, corresponding to six different ground objects ("Dirt," "Roof," "Grass," "Tree," "Road," and "Metal").

To efficiently address this sparse large-scale MOP, in this article, a global-to-local search strategy is suggested to obtain the optimal solutions. The suggested search strategy contains two stages: the global search stage and the local search stage. Specifically, two tailored solution generation strategies (asymmetric flip-based solution generation strategy and SAD-based solution repair strategy) are designed in the global search stage, with which good solutions can be found effectively in the sparse large-scale space. Then, a perturbation-based local search is developed in the second stage, which utilizes local information to further improve the quality of solutions. In addition, an endmember repetition-based solution selection strategy is proposed for both global search and local search. By using it, we can extract the endmembers with high quality. In the following, we will introduce these strategies in detail.

A. Solution Generation Strategies for Global Search

As mentioned before, the hyperspectral EE is a large-scale and sparse MOP, and the traditional evolutionary operators cannot generate good solutions effectively. To this end, two complementary solution generation strategies are proposed to generate high-quality os in the global search stage. In what follows, we will describe the main ideas and detailed procedures of these two strategies.

1) Asymmetric Flip-Based Solution Generation: In this work, since each solution (individual) is represented with the binary encoding, thus, an os can be obtained by using the 0-1 flipping operation on the parent solution (denoted as ps). To generate high-quality os in this sparse large-scale MOP, an appropriate flip probability, which includes p^{0-1} (flip from 0 to 1) and p^{1-0} (flip from 1 to 0), is designed. The determination of flip probability is based on the following three facts: 1) since the EE problem is a sparse large-scale problem, it means that in any solution, its number of 1 should be much smaller than that of 0; 2) the total number of 1 in a solution s should be equal to M; and 3) to explore the search space globally, each bit in s should have the probability of being selected as the endmember.

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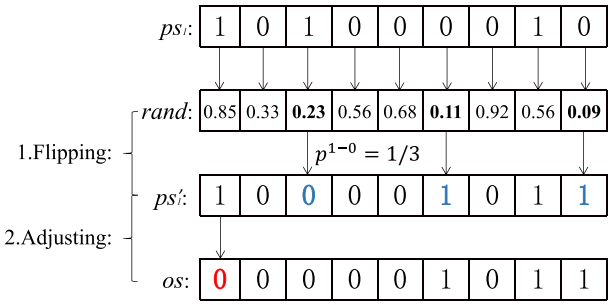

Fig. 1. Asymmetric flip-based solution generation strategy. 1: asymmetric flipping. 2: endmember adjusting.

Therefore, based on the three facts above, we define the flip probabilities p^{0-1} and p^{1-0} as

$$\begin{cases} p^{0-1} = 1/(N-M) \\ p^{1-0} = 1/M. \end{cases} \tag{6}$$

From formula (6), we can find that each bit (corresponding to a pixel in the HSI) has a certain probability of being selected as an endmember. Meanwhile, the number of endmembers after flipping can approach M as closely as possible.

With the two suggested flip probabilities, the proposed asymmetric flip-based solution generation operation can be described as follows. First, two parental solutions ps_1 and ps_2 are selected from the population through binary tournament selection. Then, the dominance relationship of ps_1 and ps_2 is compared according to two objectives defined in formula (4). If ps_1 dominates ps_2, then ps_1 is used for asymmetric flip to generate offspring, and otherwise, ps_2 is selected. If ps_1 and ps_2 are nondominated, one of them is randomly chosen. The reason underlying this selection is the intuition that a better solution is more likely to generate good offspring. Let us assume that the selected good parental solution is ps_1. For each bit i in ps_1, a random variable $\text{rand}_i \in [0, 1]$ is generated. If the value of bit i is 1 and $\text{rand}_i < p^{1-0}$, this bit will be flipped to 0, and otherwise, it remains unchanged. If the value of bit i is 0 and $\text{rand}_i < p^{0-1}$, the bit is flipped to 1, and otherwise, it remains unchanged. For all the N bits in the parent solution ps_1, the flip operation is conducted according to the bit flip probability p^{0-1} or p^{1-0}, and after that, a new solution (ps_1') is obtained.

From the descriptions above, we can find that by using the suggested flipping operations, a new solution ps_1' is generated, whose number of bits with 1 value approaches M. However, it should be admitted that this number still has the probability of not equaling M. To this end, an adjusting operator is designed to modify ps_1' to satisfy the endmembers' number M. Specifically, let us suppose that the number of bits with 1 value in ps_1' is M'. If $M' < M$, then $M - M'$ bits with 0 value in ps_1' are randomly selected and flipped to 1. If $M' > M$, then $M - M'$ bits with 1 value in ps_1' are randomly selected and flipped to 0. After the adjustment of ps_1', we set the os equaling to ps_1'. Thus, the os with M endmembers is created. The procedure of the suggested asymmetric flip-based solution generation is given in Algorithm 1.

Taking the example shown in Fig. 1 as an illustration, let us suppose that N and M equal 9 and 3, respectively. Assuming that ps_1 is the good parental solution selected from binary tournament selection, os denotes the offspring generated by ps_1 via the asymmetric flip strategy and the endmember adjusting. Specifically, for the parental solution ps_1, according to formula (6), the bit with 0 value can be flipped to 1 with the probability of 1/6, while the bit with 1 value can be flipped to 0 with the probability of 1/3. For instance, for the third bit of ps_1, its current value is 1. Based on formula (6), its p^{1-0} is 1/3, and let suppose that the third random variable rand_3 is 0.23. As $\text{rand}_3 < p^{1-0}$, thus, the third bit is flipped to 0 (marked in blue). Similarly, the flipping operation is performed on each bit of ps_1. Then, the intermediate solution ps_1' is obtained, which is shown in the third line of Fig. 1, where the bit value in blue indicates this bit is flipped. However, at this time, ps_1' has four endmembers (larger than the number of endmembers $M = 3$). Therefore, a bit with 1 value of ps_1' is randomly selected (assuming the first bit) and is flipped to 0 (marked in red). Finally, the generated solution os is shown in the last line of Fig. 1.

Algorithm 1 Asymmetric Flip-Based Solution Generation

Input: The good parent ps_1, the pixels number N, the endmembers number M.
Output: The offspring os.

```
ps'_1 ← ps_1;
for i = 1 : N do
    if ps_1(i) == 0 && rand_i(0, 1) < 1/(N − M) then
        ps'_1(i) = 1;
    else if ps_1(i) == 1 && rand_i(0, 1) < 1/M then
        ps'_1(i) = 0;
    end if
end for
M' ← Count the bits with 1 value in ps'_1;
if M' > M then
    Randomly flip (M' − M) bits in ps'_1 with 1 value to 0;
else if M' < M then
    Randomly flip (M − M') bits in ps'_1 with 0 value to 1;
end if
os ← ps'_1;
```

2) SAD-Based Solution Repair Strategy: By using the asymmetric flip-based solution generation strategy, the global search is performed on the whole space, and some latent good solutions can be obtained. However, considering that the bit flip operator above may incur the situation that more than one endmembers correspond to the same one ground object, that is to say, although the number of endmembers in a solution s equals the real number of endmembers, two or more endmembers may correspond to the same one object, which is known as the endmember repetition problem and has widely existed in many EE algorithms [35], [39].

Fig. 2 gives an example to illustrate the endmember repetition problem. Let assume that s_1 and s_2 are two solutions (see the left part of Fig. 2). The endmembers identified by s_1 and s_2 are A, B_1, C and A, B_2, C, respectively. The ground

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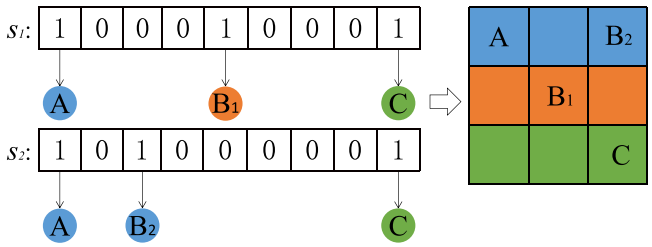

Fig. 2. Example to illustrate the endmember repetition problem.

objects are shown in the right of Fig. 2, where the same color indicates the same ground object. By comparison with the ground objects, obviously, s_2 has the problem of endmember repetition since the endmembers A and B_2 correspond to the same object (have the same color).

Thus, in the proposed method, a solution repair strategy is designed in the global search stage, which can handle the endmember repetition problem effectively. For the bad solution to be repaired, the endmembers, which correspond to the same ground object, should be identified first, and the duplicate endmembers are removed from this solution. However, a challenge lies in the fact that the ground object that each endmember corresponds to is not known in prior. To address this challenge, a learning-based repair strategy is designed, whose basic idea is from the intuition that compared with the bad solution, a good solution has less probability of endmember repetition. Thus, by learning from a good solution, the bad solution could be repaired to decrease the problem of endmember repetition. Specifically, similar to the parental solution selection in the asymmetric flip operator, two solutions are chosen through binary tournament selection. Then, the solution with higher quality is viewed as the good solution (denoted as ps_1), and the lower quality one is viewed as the bad solution to be repaired (denoted as ps_2).

Obviously, there are two crucial issues concerning the repairing process of ps_2: 1) for each learner (endmember) in the bad solution ps_2, how to determine its learning target (the corresponding endmember that to be learned from) in the good solution ps_1? and 2) how to repair the bad solution ps_2 with the learning targets of ps_1? In this regard, we implement the solution repair strategy by using the SAD metric, which is mainly divided into two steps.

Step 1: Determining learning targets in ps_1. For each endmember (denoted as e_j) in ps_2, it selects the most similar endmember (denoted as e_i) in ps_1 as its learning target. The process of measuring similarity between two endmembers is conducted by comparing the SAD values between their spectra. Thus, the learning target (e_i) in ps_1 should satisfy

$$\begin{cases} e_i = \underset{e_m \ in \ \mathrm{ps}_1}{\arg\min} \mathrm{SAD}(e_j, e_m) & (7) \\ \mathrm{SAD}(e_j, e_m) = \cos^{-1}\left(\frac{e_j \cdot e_m}{\|e_j\| \cdot \|e_m\|}\right). & (8) \end{cases}$$

Note that the SAD value is often used to measure the difference between two spectra. When the SAD value between two endmembers is small, it means that the two endmembers are similar and vice versa. Through this way, each endmember in ps_2 can find its learning target in ps_1.

Algorithm 2 SAD-Based Solution Repair

Input: The good parent ps_1, the bad parent ps_2, the endmembers number M.
Output: The offspring os.

$\{e_m\}_{m=1}^{M} \leftarrow$ Get the endmember spectra set in ps_1;
$\{e_j\}_{j=1}^{M} \leftarrow$ Get the endmember spectra set in ps_2;
for each e_j in ps_2 **do**
  $e_i \leftarrow$ Calculate the arg min SAD value of e_j and e_m by (7);
  Record the SAD values of e_i and e_j;
end for
if one-to-one relationship is satisfied **then**
  // Case 1:
  $e_p \leftarrow$ Select the endmember with the largest SAD value from $\{e_j\}_{i=1}^{M}$ in ps_2;
  $e_q \leftarrow$ Get the learning target of e_p;
  $os \leftarrow$ The endmember set that after repairing is $\{e_j\}_{j=1}^{M} \setminus e_p \cup e_q$;
else
  // Case 2:
  for each e_m in ps_1 **do**
    if e_m is the learning target **then**
      $\{e_{sl}\} \leftarrow$ Get the endmembers in ps_2 which correspond to e_m;
      $e_{rl} \leftarrow$ Get the endmember with the least SAD value from $\{e_{sl}\}$;
      Store e_{rl} in a endmembers set $\{e_{rl}\}$;
    end if
  end for
  $\{e_{nl}\} \leftarrow$ Get the endmembers which are not the learning target in ps_1;
  $os \leftarrow$ The endmember set that after repairing is $\{e_{rl}\} \cup \{e_{nl}\}$;
end if

Step 2: Repairing the bad solution ps_2. When the learning targets of the endmembers in ps_2 are determined, it can be repaired by learning from the good solution ps_1. According to the relationship between learners and learning targets, there are two cases to be considered.

1) *Case 1 (One-to-One Relationship:* Each endmember in ps_1 is the learning target for only one endmember in ps_2. In this case, the endmember in ps_2 whose SAD value with its learning target is the largest will be replaced by the corresponding learning target in ps_1. The reason lies in the fact that for ps_2, this learning from ps_1 can make ps_2 more approximate ps_1, and the endmember repetition in ps_2 can be reduced effectively.
2) *Case 2 (Multiple-to-One Relationship):* When two or more endmembers in ps_2 take the same endmember in ps_1 as the learning target, these endmembers in ps_2 are likely to be located in the same ground object region. In this case, for these endmembers in ps_2, only the endmember whose SAD value with its learning target is the smallest will be reserved, and the other endmembers will be deleted. Meanwhile, the endmembers, which are not taken as the learning target for any endmember in ps_1, are added into the endmember set in ps_2.

Through the abovementioned SAD-based solution repair strategy, we can generate a new solution, whose endmember

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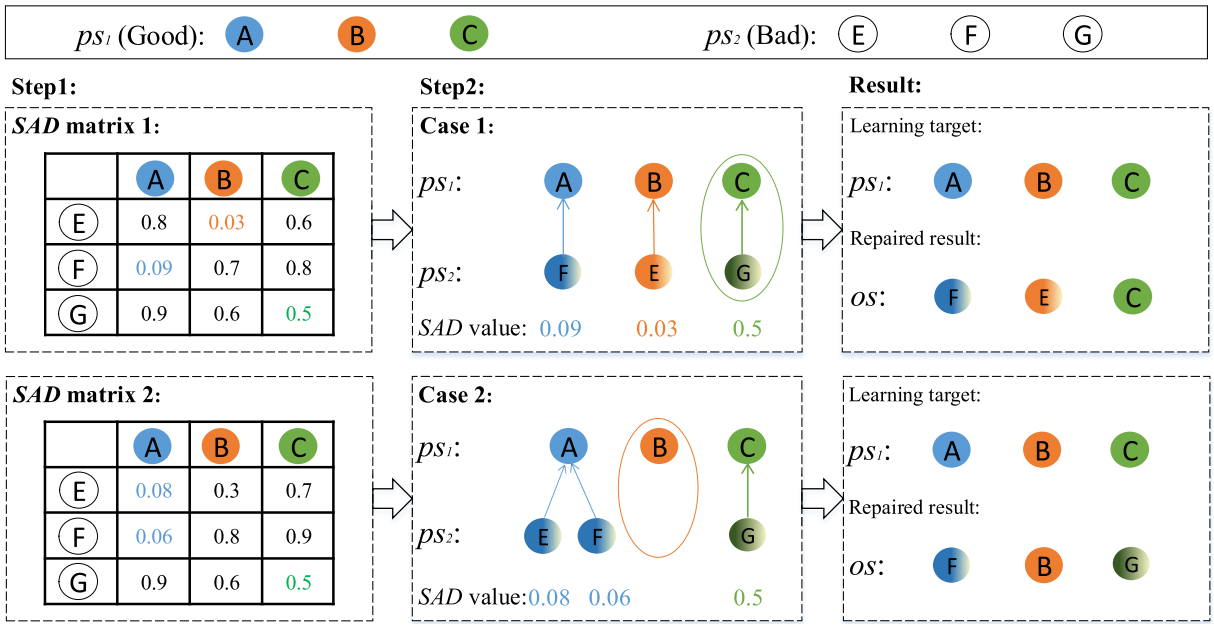

Fig. 3. Illustrative example of the suggested SAD-based solution repair strategy.

repetition problem is alleviated greatly. The detailed procedure of SAD-based solution repair strategy is described in Algorithm 2.

Fig. 3 further gives an illustrative example for the suggested SAD-based solution repair strategy, where circles with similar colors represent the pixels belonging to the same ground object. For the selected solution pair ps_1 and ps_2, let us assume that ps_1 is a good solution and ps_2 is a bad solution to be repaired. In the first step, the SAD values among any ps_1's endmembers and ps_2's endmembers are calculated and shown in the SAD matrix. Specifically, from the SAD matrix 1 (see the top-left part of Fig. 3), according to formulas (7) and (8), A is the learning target of F, while B and C are the learning targets of E and G, respectively. Thus, this situation corresponds to Case 1: one-to-one relationship (see the top middle part of Fig. 3). Then, in the repairing process, the endmember G in ps_2 is replaced by C since the SAD value between G and C is the largest. Thus, ps_2 after the repairing operator is composed of F, E, and C (see the top right of Fig. 3). The bottom of Fig. 3 presents another case. From SAD matrix 2, it can be derived that A is the learning target for both E and F; however, B is not the learning target for any endmember in ps_2. Thus, this situation refers to Case 2: multiple-to-one relationship. Considering that the SAD value of E (0.08) is larger than that of F (0.06), and thus, E is removed. Meanwhile, the endmember B in ps_1 is not the learning target of any endmember in ps_2 and is added to the repaired endmembers in ps_2. Finally, the repaired endmembers in ps_2 are composed of F, B, and G.

B. Solution Generation Strategy for Local Search

Through the suggested global search, good solutions that satisfied the constraint can be found in the sparse large-scale space. To further enhance the quality of the final solutions, in the proposed GL-EA, a local search strategy is also designed and the idea is inspired by the following intuition. Due to the spatial consistency in the HSI, the surrounding pixels adjacent to an endmember position are likely to belong to the same ground object. Therefore, it is possible that better endmember position may be located near current endmember position. Fig. 4 gives an example to illustrate this intuition. As shown in Fig. 4(a) and (b), a better endmember position M_i' for the ground object of "tree" is adjacent to current endmember position M_i. Obviously, in M_i', trees occupy a larger area than in M_i. In addition, the spectral curve also shows that M_i' are more closer to the reference than that of M_i [see Fig. 4(c)].

Based on the intuition above, we propose a perturbation-based local search in the second stage. Considering that if all the positions near the current endmember position are considered in the local search, the evaluation costs will increase greatly. Therefore, in our work, at each generation, for each solution, only one endmember is chosen randomly for local search. Meanwhile, only one position adjacent to the selected endmember position is perturbed. Formally, the local search is conducted in the following way. First, a solution ps_1 is selected from the population through binary tournament selection. For a randomly selected endmember position M_i in ps_1, two random numbers d_x and d_y located in $[-r, r]$ (r is a parameter that controls the search range) are generated. Afterward, a new endmember position is yielded by

$$\begin{cases} M_i'(x) = M_i(x) + d_x \\ M_i'(y) = M_i(y) + d_y \end{cases} \tag{9}$$

where $M_i(x)$ and $M_i'(x)$ represent the horizontal components of the original and new endmember positions, respectively, while $M_i(y)$ and $M_i'(y)$ correspond to the vertical components. After that, a new offspring (os), which contains M_i' and excludes M_i, is generated. Algorithm 3 gives the detailed procedure of the suggested perturbation-based local search.

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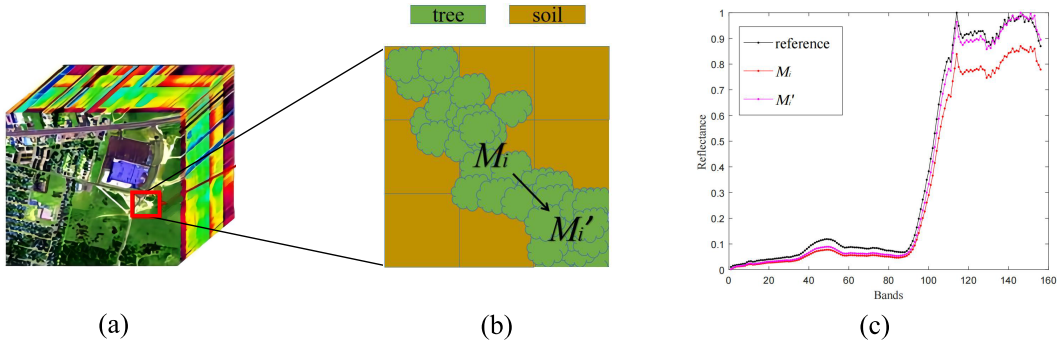

Fig. 4. Example to illustrate the intuition of the suggested local search. (a) HSI. (b) 3 × 3 image region including "trees" and "soil" ground objects. (c) Reference spectra of "tree" endmember and the spectra at M and M_i'.

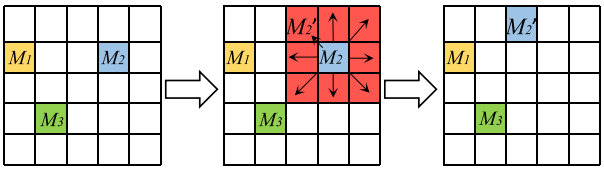

Fig. 5. Example to illustrate the procedure of the suggested perturbation-based local search. (a) Three endmembers in a solution ps_1. (b) Possible new position of endmember M_2. (c) Final new position of endmember M_2.

To illustrate how M_i is changed to M_i', an example is given in Fig. 5. As shown in Fig. 5(a), M_1, M_2, and M_3 are three endmember positions in a solution ps_1. Assume that r is set to 1 in this example, and thus, the endmember position is perturbed in the range of $[-1, 1]$. Let us suppose that M_2 is randomly selected to change its position in the search range. According to formula (9), the new possible endmember positions are marked in red [see Fig. 5(b)]. From these new positions, an endmember position is randomly selected as M_2' [see the top left of M_2 in Fig. 5(b)]. The new M_2' combined with M_1 and M_3 to form an offspring individual os [see Fig. 5(c)].

Algorithm 3 Perturbation-Based Local Search

Input: The good parent ps_1, the endmembers number M, the parameter r.

Output: The offspring os.

1: $\{pos_i\}_{i=1}^{M} \leftarrow$ Get the endmember position set in ps_1;
2: $k \leftarrow randint([1, M])$; // one bit is randomly selected;
3: $d_x \leftarrow randint([-r, r])$;
4: $d_y \leftarrow randint([-r, r])$;
5: $pos_k^{new} \leftarrow$ Calculate from pos_k by (9);
6: $os \leftarrow$ Convert the endmember position set $\{pos_i\}_{i=1}^{M} \setminus pos_k \cup pos_k^{new}$ into the solution os;

C. Endmember Repetition-Based Solution Selection

Through the above global-to-local search, the os with high quality can be generated. However, it is well known that in the literature of evolutionary computation beside solution generation, the solution selection strategy also has a great effect on the quality of final solutions. To this end, in this work, the nondominated sorting suggested in NSGA-II [47] is adopted to assign individuals to different Pareto layers. However, for the individuals in the same Pareto layer, an endmember repetition-based solution selection strategy is designed to replace the crowding distance, which are applied in both global and local search stages and can select solutions with higher quality.

The basic idea of this designed strategy is inspired by the fact that the endmembers in a good solution should be different from each other. Therefore, if the endmembers in a solution are not similar to each other, this solution is regarded as a good one; otherwise, it is not viewed as a good one. Thus, the endmember repetition in a solution can be taken as the metric for the determination. Specifically, for a solution s, we define its minSAD (s) metric as the minimum of the SAD values for all the combinations of any two different endmembers inside s. Therefore, in the designed solution selection strategy, for the solutions in the same layer, we sort these solutions according to their minSAD (s) values, and the solutions with higher minSAD (s) values will be selected. Fig. 6 gives an example of the calculation of the minSAD (s). Let us assume that there are m solutions in the same layers [see Fig. 6(a)], from which m' $(m' \leq m)$ solutions need be chosen for the next generation. Thus, for each solution s_i (e.g., $s_1, \ldots, s_m$), we calculate its minSAD (s_i). We take solution s_1 as an example [see the top part of Fig. 6(b)]. Since it has three endmembers (A, B, and C), its endmember combinations are $\{A, B\}$, $\{B, C\}$, and $\{A, C\}$, respectively. By using formula (8), we can obtain their corresponding SAD values (assume that they are 0.4, 0.8, and 0.02). Thus, minSAD for s_1 is 0.02. Similarly, we can calculate the minSAD (s_i) for $s_1, \ldots, s_m$ and the results are shown in Fig. 6(c). Then, according to their minSAD values, we sort $s_1, \ldots, s_m$ in a descending order, and the top m' solutions are chosen for the next generation [see Fig. 6(d)]. The detailed procedure of endmember repetition-based solution selection is depicted in Algorithm 4.

D. General Framework of GL-EA

Based on the global-to-local solution generation strategy and the designed endmember repetition-based solution selection strategy, the high-quality endmembers can be extracted. To be specific, the general framework of GL-EA is presented in Algorithm 5, which can be mainly divided into three stages: the initialization stage, the global search stage, and the local search stage. In the initialization stage (Lines 1–3),

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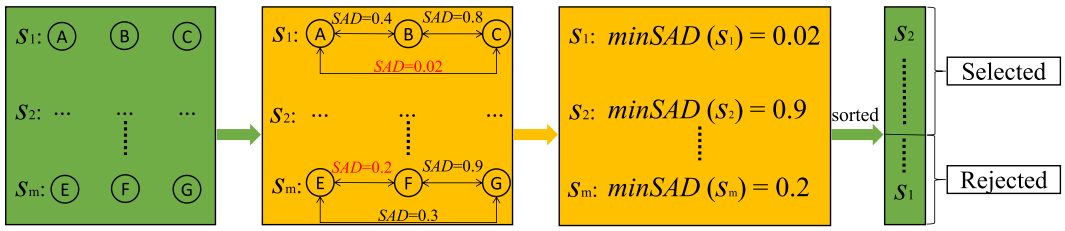

Fig. 6. Example to illustrate the endmember repetition-based solution selection strategy. (a) m solutions in the same layers. (b) Calculating the SAD value of each endmember combination in a solution. (c) Calculating the minSAD value of each solution. (d) Sorting m solutions in a descending order according to their minSAD values.

Algorithm 4 Endmember Repetition-Based Solution Selection

Input: The population size NP, the current iteration t, the population of parent $\boldsymbol{P}_t$, the population of offspring $\boldsymbol{O}_t$, the endmembers number M.
Output: New population $\boldsymbol{P}_{t+1}$.
1: $\boldsymbol{P}_c \leftarrow \boldsymbol{P}_t \cup \boldsymbol{O}_t$;
2: $F_n \leftarrow$ Determine the Pareto layer to be selected in P_c according to non-dominated sorting suggested in NSGA-II;
3: $P1_t \leftarrow$ Get the individuals in F_1 to F_{n-1};
4: $N' \leftarrow$ Count the number of individuals in F_1 to F_{n-1};
5: $\{s_i\}_{i=1}^{m} \leftarrow$ Get all individuals in the F_n layer;
6: **for** $i = 1 : m$ **do**
7: $\{e_j\}_{j=1}^{M} \leftarrow$ Get the endmember spectra set in s_i;
8: $minSAD\ (s_i) \leftarrow$ Minimize the $SAD(e_j, e_{j'})$ while $j \neq j'$ and $e_j, e_{j'} \in \{e_j\}_{j=1}^{M}$;
9: **end for**
10: $P2_t \leftarrow$ Get the top $(NP - N')$ individuals from $\{s_i\}_{i=1}^{m}$ sorted by $minSAD\ (s_i)$ in a descending order;
11: $\boldsymbol{P}_{t+1} \leftarrow P1_t \cup P2_t$;

the population P_0 is generated through random initialization. Afterward, the initial population is fed to the global search stage (Lines 4–18). In the global search stage, there are two branches, which are conducted in a random way. One branch is the suggested asymmetric flip-based solution generation strategy (Line 11), which is utilized to explore the search space as broadly as possible. Another branch is the suggested SAD-based solution repair strategy (Line 13), which is adopted to repair the global solutions for better quality. Then, the nondominated sorting and the designed endmember repetition-based solution selection strategy (Line 16) are used to select individuals for the next generation. The above global search stage iterates for maxgen $* K$ generations ($0 \leq K \leq 1$), and then, the proposed GL-EA comes to the local search stage (Lines 19–29). In this stage, the suggested perturbation-based local search (Line 25) is utilized to search the better solutions around current endmember positions. Then, the environmental selection strategy that is the same as the global search stage is used to select individuals for next generation. The local search repeats for maxgen $* (1 - K)$ iterations. Finally, the final PF is adopted as the output.

Algorithm 5 General Framework of GL-EA

Input: The population size NP, the maximum generation $maxgen$, the endmembers number M, the pixels number N, the parameters r and K.
Output: Pareto set PF.
1: Stage 1: initialization;
2: $t = 0$;
3: Randomly initialize the population of parent: $\boldsymbol{P}_0 = \{s_{1,0}, s_{2,0}, \ldots, s_{NP,0}\}$, the population of offspring: $\boldsymbol{O}_0 = NULL$;
4: Stage 2: global search;
5: **while** $t \leq maxgen * K$ **do**
6: **for** $i = 1 : NP$ **do**
7: $s_1' \leftarrow$ Select by binary tournament selection;
8: $s_2' \leftarrow$ Select by binary tournament selection;
9: $\{\mathrm{ps}_1(good), \mathrm{ps}_2(bad)\} \leftarrow$ Get by comparing the dominance of $\{s_1', s_2'\}$;
10: **if** $rand() < 0.5$ **then**
11: $\boldsymbol{O}_{i,t} \leftarrow$ Asymmetric flip based solution generation (ps_1, N, M); // see **Algorithm 1**;
12: **else**
13: $\boldsymbol{O}_{i,t} \leftarrow$ SAD based solution repair $(\mathrm{ps}_1, \mathrm{ps}_2, M)$; // see **Algorithm 2**;
14: **end if**
15: **end for**
16: $\boldsymbol{P}_{t+1} \leftarrow$ Endmember repetition based solution selection $(NP, t, \boldsymbol{P}_t, \boldsymbol{O}_t, M)$; // see **Algorithm 4**;
17: t++;
18: **end while**
19: Stage 3: local search;
20: **while** $maxgen * K < t \leq maxgen$ **do**
21: **for** $i = 1 : NP$ **do**
22: $s_1' \leftarrow$ Select by binary tournament selection;
23: $s_2' \leftarrow$ Select by binary tournament selection;
24: $\{\mathrm{ps}_1(good), \mathrm{ps}_2(bad)\} \leftarrow$ Get by compare the dominance of $\{s_1', s_2'\}$;
25: $\boldsymbol{O}_{i,t} \leftarrow$ Perturbation based local search (ps_1, M, r); // see **Algorithm 3**;
26: **end for**
27: $\boldsymbol{P}_{t+1} \leftarrow$ Endmember repetition based solution selection $(NP, t, \boldsymbol{P}_t, \boldsymbol{O}_t, M)$; // see **Algorithm 4**;
28: t++;
29: **end while**
30: PF $\leftarrow$ Get from $\boldsymbol{P}_t$;

E. Time Complexity Analysis of GL-EA

In this section, we give the time complexity analysis of GL-EA. Specifically, according to the general framework

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of GL-EA (See Algorithm 5), the proposed method mainly includes three stages. The first stage is the initialization (Lines 1–3). In this stage, NP individuals in the population are randomly initialized. Thus, the computational complexity of the first stage is $O(NP)$. The second stage is the global search (Lines 4–18). In its inner iteration (Lines 6–15), for the NP individuals (Line 6), each one has three suboperations.

1) The binary tournament selections and comparing the dominance (Lines 7–9), whose computational cost is $O(1)$.
2) The asymmetric flip-based solution generation (Lines 10–12). According to its details in Algorithm 1, its computational complexity is $O(N)$, where N is the number of pixels.
3) The SAD-based solution repair (Lines 13 and 14). Its details are described in Algorithm 2. According to Algorithm 2, the computational complexity of SAD-based solution repair is $O(M^2)$, where M is the number of endmembers.

The above three suboperations repeat for NP individuals, and their total computational complexity is $O(NP \cdot (N + M^2))$. Then, Line 16 is the suggested endmember repetition-based solution selection. According to its implement (see Algorithm 4), its computational complexity is $O(NP^2 + NP \cdot M^2)$. Combining Lines 6–17, the computational complexity in each global search evolve is $O(NP^2 + NP \cdot (N+M^2))$. Since the global search repeats for maxgen evolves, the whole computational cost in the global search stage (the second stage) is $O(\text{maxgen} \cdot (NP^2 + NP \cdot (N + M^2)))$. The third stage is the local search phase (Lines 19–29). The whole procedure of local search is similar to that of global search, except that in its inner iterations, it performs the perturbation-based local search (the details can be found in Algorithm 3). According to Algorithm 3, its computational cost is $O(N)$. Thus, the computational complexity of the local search is $O(\text{maxgen} \cdot (NP \cdot N + NP^2 + NP \cdot M^2))$, which is $O(\text{maxgen} \cdot (NP^2 + NP \cdot (N + M^2)))$. Finally, we combine stages 1–3, we can obtain that the total computational complexity of the proposed GL-EA is $O(\text{maxgen} \cdot (NP^2 + NP \cdot (N + M^2)))$.

IV. Experimental Results

In this section, a series of experiments is conducted to verify the effectiveness of GL-EA. Specifically, we first present the experimental settings, which include comparison algorithms, benchmark datasets, and the parameter settings for all the algorithms. Then, the experimental results are reported and analyzed from different aspects. Finally, we validate the effectiveness of the suggested strategies in GL-EA.

A. Experimental Settings

1) Comparison Algorithms: The proposed GL-EA is compared with five representative EE algorithms, namely, N-FINDR [18], DPSO [35], MODPSO [38], IMODPSO [39], and $(\mu + \lambda)$MODE [41]. Among them, N-FINDR is a traditional non-EA-based EE algorithm. DPSO belongs to the single-objective EA-based EE method, and other three algorithms are MOEA-based EE algorithms, which are the same as ours. Specifically, N-FINDR obtains endmembers via seeking for a simplex with the maximum volume. In this method, M pixels are first initialized as endmembers and the corresponding volume is calculated. Then, each endmember is replaced by other pixels in turn and the volume is evaluated. Finally, the pixels that achieve the largest volume are considered as endmembers. DPSO is based on the discrete PSO algorithm, where the RMSE value is used as the objective function, and new velocity and position update methods are designed for generating particles with good quality.

Three state-of-the-art MOEA-based EE algorithms, called MODPSO, IMODPSO, and $(\mu + \lambda)$MODE, are selected as the comparison methods since they are recently suggested methods and have shown promising performance in tackling the EE problem. To be specific, MODPSO is a multiobjective discrete PSO algorithm, where RMSE and the volume inverse are the optimization objectives. IMODPSO features the restart mechanism strategy, which can remove the duplicate particle. In addition, an adaptive archiving mechanism is also designed in IMODPSO, which increases the population's diversity. $(\mu + \lambda)$MODE is a recently suggested MOEA for the EE task. In $(\mu + \lambda)$MODE, two different DE operators for solution generation are designed for two subpopulations, with which $(\mu + \lambda)$MODE extracts endmembers with higher quality.

2) Datasets: Three classic datasets (Samson, Jasper Ridge, and Urban)[1] with about 9000–30 000 pixels are adopted as the benchmark datasets. These HSIs contain different bands, pixels, and ground objects, which are often used in testing the performance of different EE algorithms [38], [39], [40], [41].

The spatial dimension of Samson dataset is 95 × 95, and each pixel contains 156 spectral bands with wavelength ranging from 401 to 889 nm. The endmembers existing in the Samson dataset are "1 Soil," "2 Tree," and "3 Water." The spatial dimension of Jasper Ridge dataset is 100 × 100, and each pixel contains 198 spectral bands with the wavelength ranging from 380 to 2500 nm. There are four endmembers in the Jasper Ridge dataset: "1 Tree," "2 Dirt," "3 Water," and "4 Road." The spatial dimension of Urban dataset is 307 × 307 and each pixel contains 162 spectral bands with the wavelength ranging from 400 to 2500 nm. Six endmembers exist in the Urban dataset: "1 Dirt," "2 Roof," "3 Grass," "4 Tree," "5 Road," and "6 Metal." The pseudo-color images of these datasets are shown in Fig. 7.

3) Parameter Settings: For the five comparison algorithms, the recommended parameter settings suggested in their original papers are adopted. To ensure the fairness in comparison, for all the EAs, the population size is specified to 30, and the maximum number of generations is fixed to 200. For our proposed GL-EA, the parameter K that used to determine the period of global and local search is set to 0.8. In addition, in the local search stage, r for perturbation-based local search is set to 5 for each dimension. All the experiments are performed on a PC with Intel[2] Core[3]

[1]The datasets can be downloaded from https://rslab.ut.ac.ir/data
[2]Registered trademark.
[3]Trademarked.

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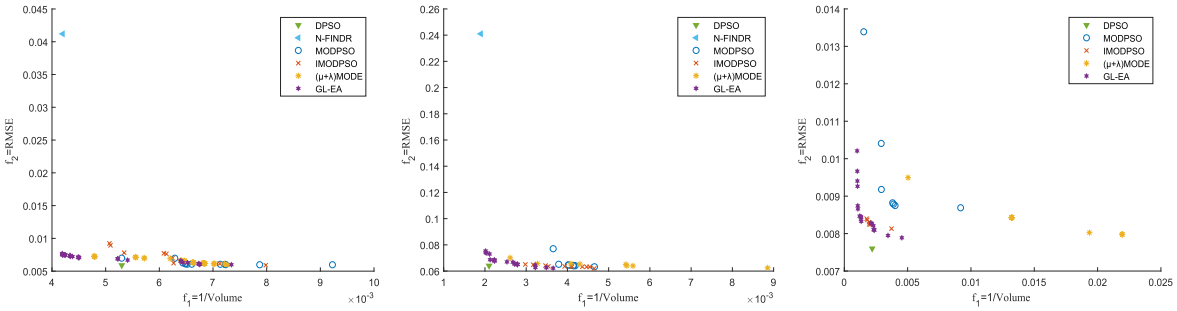
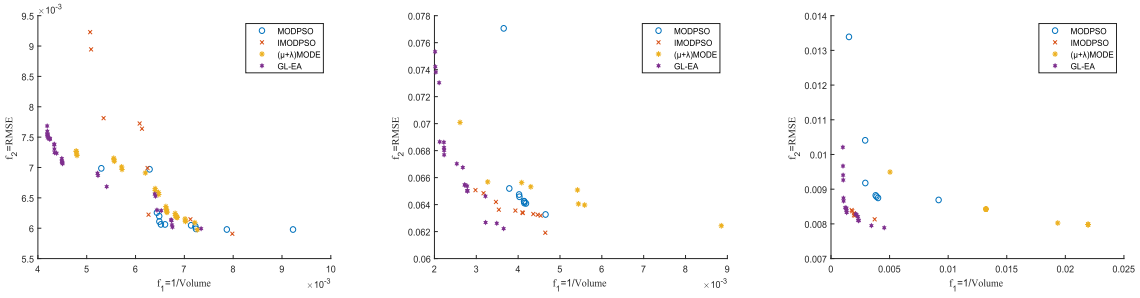
(a) (b) (c)

Fig. 7. Pseudo-color images of three datasets. (a) Samson. (b) Jasper Ridge. (c) Urban.



(a) (b) (c)

Fig. 8. Comparison results of all the methods on the three datasets. (a) Samson dataset. (b) Jasper Ridge dataset. (c) Urban dataset.



(a) (b) (c)

Fig. 9. PF results of four MOEA-based algorithms on the three datasets. (a) Samson dataset. (b) Jasper Ridge dataset. (c) Urban dataset.

i5-9500 CPU@3.00GHz, 8GB RAM, and the Windows 10 operating system.

B. EE Results and Analysis

1) Overall Performance: First, the overall performance of different methods is verified. To this end, we report the comparison results of six algorithms in terms of the RMSE value and the volume inverse, which are two popular metrics in the EE area. Specifically, for the four MOEAs (MODPSO, IMODPSO, $(\mu + \lambda)$MODE, and the proposed GL-EA), we adopt their PFs since the RMSE value and the volume inverse are their two optimized objectives, while the two single-objective algorithms, N-FINDR and DPSO, only optimize one objective. To make the comparison fair, we adopt the common practice in EE [38], [41] that for N-FINDR and DPSO, each algorithm is conducted twice. The RMSE value and the volume inverse are taken as the optimization objectives for each time. Then, the two results of N-FINDR or DPSO are taken for comparison. The detailed comparison results of different algorithms are shown in Fig. 8. Note that for N-FINDR on the Urban dataset, a total of $6 \times 307 \times 307$ evaluations are required to optimize the RMSE objective with Ncls, which causes the result to be obtained in 100 h. Hence, in Fig. 8(c), there is no result of N-FINDR. From the results in Fig. 8, when measured by two metrics, the proposed GL-EA and DPSO outperform other algorithms in most cases. The reason that DPSO has a better performance is attributed to the fact that this result is obtained by combining its twice running, which optimizes RMSE and 1/volume. Different from DPSO that has to run twice, the proposed GL-EA achieves the better performance by only running once. Meanwhile, the results in Fig. 8 also reveal that GL-EA has several better solutions, while DPSO only has one better solution.

To further compare the performance of four MOEAs, we also present the results of these MOEAs in an emphasized

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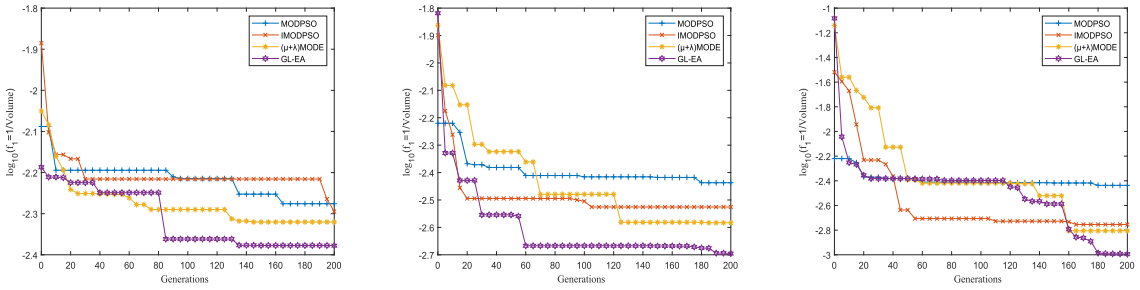

Fig. 10. Convergence curves of different MOEAs in terms of 1/Volume on the thre datasets. (a) Samson. (b) Jasper Ridge. (c) Urban.

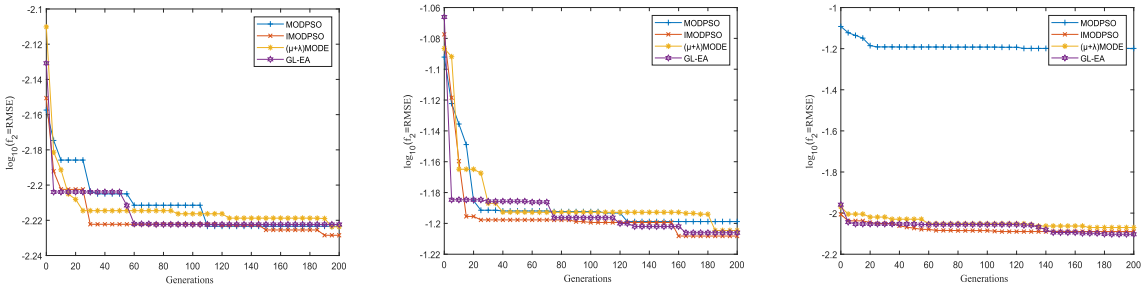

Fig. 11. Convergence curves of different MOEAs in terms of RMSE value on the three datasets. (a) Samson. (b) Jasper Ridge. (c) Urban.

TABLE I
OBJECTIVE RESULTS OF ALL THE MOEA-BASED ALGORITHMS ON THE THREE DATASETS

Data sets	Samson		Jasper Ridge		Urban	
Algorithm	$1/Volume$	$RMSE$	$1/Volume$	$RMSE$	$1/Volume$	$RMSE$
MODPSO	5.30E-03	6.99E-03	3.66E-03	7.71E-02	1.53E-03	1.34E-02
	7.87E-03	5.97E-03	4.66E-03	6.33E-02	9.18E-03	8.69E-03
IMODPSO	5.07E-03	9.22E-03	2.98E-03	6.51E-02	1.76E-03	8.40E-03
	7.98E-03	**5.90E-03**	4.65E-03	**6.19E-02**	3.73E-03	8.13E-03
(μ+λ)MODE	4.79E-03	7.21E-03	3.74E-03	6.65E-02	5.04E-03	9.49E-03
	7.21E-03	6.09E-03	4.48E-03	6.31E-02	2.19E-02	7.98E-03
GL-EA	**4.19E-03**	7.55E-03	**2.02E-03**	7.53E-02	**1.01E-03**	1.02E-02
	7.34E-03	5.99E-03	3.49E-03	6.26E-02	4.54E-03	**7.89E-03**

TABLE II
HV VALUES OF ALL THE MOEA-BASED METHODS ON THE THREE DATASETS

Data sets	Samson	Jasper Ridge	Urban
MODPSO	0.2264	0.1847	0.5669
IMODPSO	0.2245	0.2127	0.5964
(μ+λ)MODE	0.2412	0.2041	0.5194
GL-EA	**0.2639**	**0.2330**	**0.6204**

way, which is shown in Fig. 9. From Fig. 9, we can find that compared with other three MOEAs (MODPSO, IMODPSO, and (μ + λ)MODE) for the EE problem, the proposed GL-EA has a better performance in terms of both RMSE and volume inverse. Table I also gives the solutions of four MOEAs with the minimal RMSE and volume inverse values on the benchmarks, from which we observe that on the three datasets with six statistical points, the proposed GL-EA achieves the minimum value on 4/6 and is the second, and the third best on the remained two points. The statistics in Table I have verified the effectiveness of GL-EA again.

2) Convergence Speed: The superior performance of GL-EA is attributed to the suggested global-to-local evolutionary strategy, which not only ensures that the proposed method achieves better solutions but also enables that GL-EA have faster convergence speed. To verify the fact, in this section, we present the convergence curves of different MOEA-based EE algorithms on the three datasets. The results are plotted in terms of 1/volume (log) and RMSE (log), which are shown in Figs. 10 and 11, respectively.

From Fig. 10, we can find that when measured by 1/volume, GL-EA has much faster convergence speed than MODPSO, IMODPSO, and (μ + λ)MODE. Meanwhile, Fig. 11 also shows that when measured by the RMSE, GL-EA has faster convergence than MODPSO, (μ + λ) MODE, and has similar speed to IMODPSO. The results in

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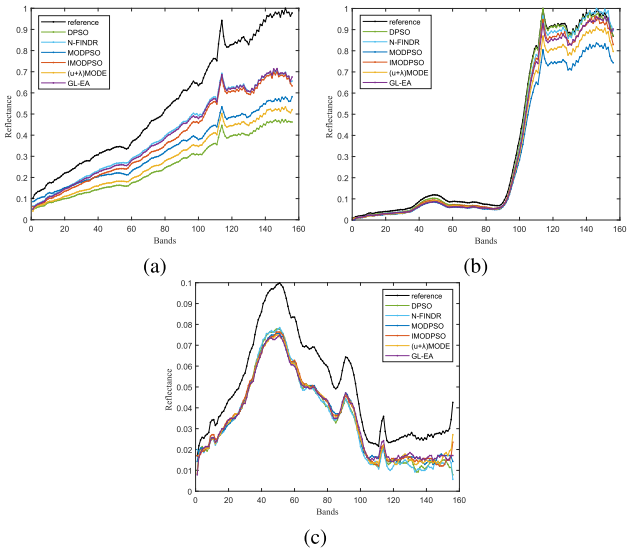

Fig. 12. Endmembers spectra of different algorithms on the Samson dataset. (a) Soil. (b) Tree. (c) Water.

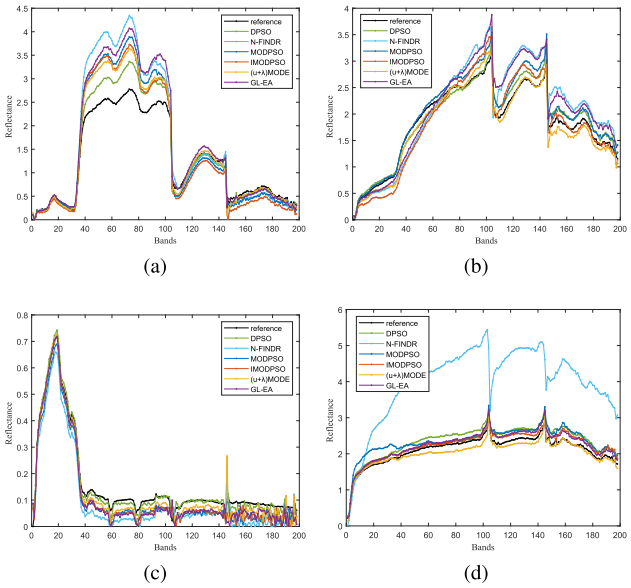

Fig. 13. Endmembers spectra of different algorithms on the Jasper Ridge dataset. (a) Tree. (b) Dirt. (c) Water. (d) Road.

Figs. 10 and 11 have verified the fast convergence speed of the GL-EA.

3) HV Values: Furthermore, since our GL-EA is an MOEA, thus, in this section, we compare four MOEA-based EE algorithms from the aspect of MOP. Specifically, we adopt HV value as the metric, which is a popular metric to measure both the diversity and convergence of different MOEAs [58], [59]. The larger HV value means the better performance of the MOEA. The HV results of four MOEA-based EE algorithms are listed in Table II. It can be observed from the table that GL-EA achieves the highest HV values on all the three datasets. This fact confirms that when evaluated from MOP, the proposed GL-EA still has better diversity and faster convergence.

C. SAD Values' Comparison and Analysis

In this section, we further validate the competitiveness of the proposed GL-EA by comparing the spectral similarity between the extracted endmembers and its corresponding ground objects. To this end, the SAD value is adopted as the metirc, which can be calculated by formula (8). A lower SAD value means that the extracted endmember is more closer to the ground object. Considering that four MOEA-based EE algorithms (MODPSO, IMODPSO, $(\mu + \lambda)$MODE, and the proposed GL-EA) have several solutions in one run, we select the solution with the lowest SAD value for each MOEA. The SAD value results of all six EE algorithms on three datasets are shown in Tables III–V. Note that in all the tables for each dataset, the best (lowest) SAD value is marked with boldface. The symbol "-" in the tables denotes the algorithm that does not extract this endmember.

From Tables III–V, we can find that four MOEAs have lower SAD values than the two single-objective algorithms in most cases. Especially on Urban dataset, both N-FINDR and DPSO cannot extract some endmembers (e.g., "Metal"). This fact demonstrates the superiority of MOEAs for solving the problem of EE.

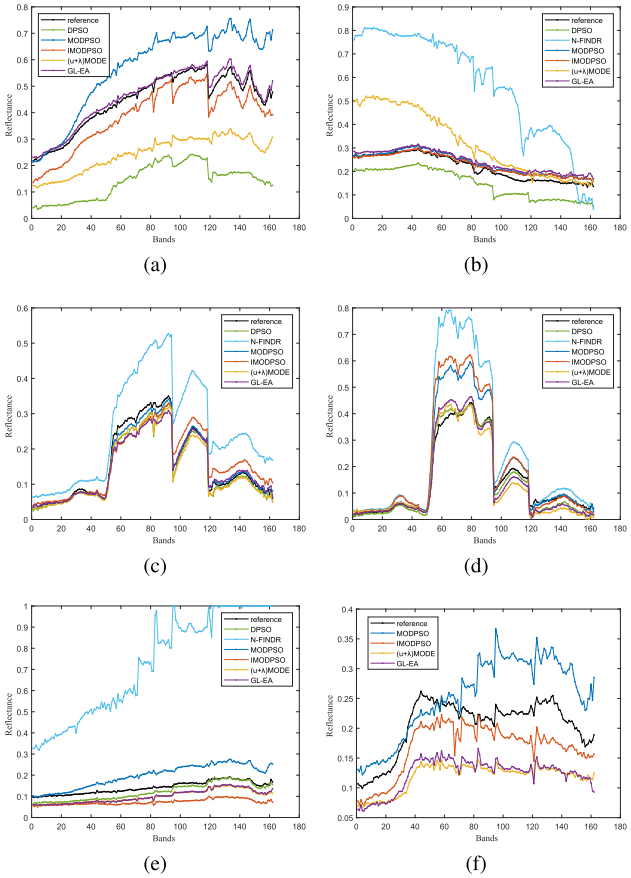

Fig. 14. Endmembers spectra of different algorithms on the Urban dataset. (a) Dirt. (b) Roof. (c) Grass. (d) Tree. (e) Road. (f) Metal.

By taking a closer look at four MOEAs on three datasets, we have the following observations. First, from Table III, we can find that on the Samson dataset, the proposed GL-EA obtains the best SAD values for all the endmembers ("Soil," "Tree," and "Water"), which is much better than other three MOEA-based EE algorithms (MODPSO, IMODPSO,

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TABLE III
MINIMUM SAD RESULTS OF DIFFERENT ALGORITHMS ON THE SAMSON DATASET

	DPSO	N-FINDR	MODPSO	IMODPSO	$(\mu+\lambda)$MODE	GL-EA
1-Soil	0.0257	0.0402	0.0098	0.0111	0.0093	**0.0079**
2-Tree	0.0219	0.0407	0.0293	0.0173	0.0354	**0.0160**
3-Water	0.0648	0.1224	0.0417	0.0555	0.0556	**0.0303**

TABLE IV
MINIMUM SAD RESULTS OF DIFFERENT ALGORITHMS ON THE JASPER RIDGE DATASET

	DPSO	N-FINDR	MODPSO	IMODPSO	$(\mu+\lambda)$MODE	GL-EA
1-Tree	0.15049	0.1353	**0.0367**	0.0639	0.0415	0.0539
2-Dirt	0.0567	0.1336	0.0354	**0.0294**	0.0493	0.0302
3-Water	0.1654	0.2543	0.0793	0.189	0.0750	**0.0699**
4-Road	0.0349	0.1069	0.0216	**0.0197**	0.0248	**0.0197**

TABLE V
MINIMUM SAD RESULTS OF DIFFERENT ALGORITHMS ON THE URBAN DATASET

	DPSO	N-FINDR	MODPSO	IMODPSO	$(\mu+\lambda)$MODE	GL-EA
1-Dirt	0.0420	-	0.0594	0.0225	0.0627	**0.0202**
2-Roof	0.1015	0.1740	0.0545	0.0420	0.0336	**0.0302**
3-Grass	0.0630	0.1160	**0.0314**	0.0421	0.0385	0.0415
4-Tree	0.1343	0.1227	0.0815	0.0468	0.0701	**0.0695**
5-Road	0.0429	0.1480	**0.0262**	0.0697	0.0552	0.0504
6-Metal	-	-	0.1904	0.0804	0.0653	**0.0619**

and $(\mu + \lambda)$MODE). Second, on the Jasper Ridge dataset, our GL-EA can achieve the best SAD values for the "Water" and "Road" endmembers, while MODPSO achieves the best SAD value for the "Tree" endmember, and IMODPSO gets the best SAD value for the "Dirt" and "Road" endmembers. These statistics in Table IV reveal that the proposed GL-EA and IMODPSO have better performance on the Jasper Ridge dataset. Third, on the Urban dataset, GL-EA can obtain the best SAD values for 4/6 endmembers ("Dirt," "Roof," "Tree," and "Metal"). The MODPSO obtains the best SAD values for the remained two endmembers ("Grass" and "Road"). From the three observations, we can conclude that although different EE algorithms have different performances, our GL-EA always extracts endmembers with higher quality.

In addition, Figs. 12–14 present the spectra for the extracted endmembers of six EE algorithms on three datasets. Note that in each figure, the black curve denotes the corresponding reference endmember. It can be found that different algorithms have different spectra, and however, the proposed GL-EA has more similar spectra to the reference endmembers in most cases when compared with other EE algorithms.

Combining all the experimental results in Sections IV-B and IV-C, we can conclude that compared with the state of the arts, our GL-EA is a competitive EE algorithm, which can extract the endmembers with higher quality.

D. Effectiveness of the Proposed Strategies in GL-EA

All the above experimental results demonstrate the superiority of the proposed GL-EA. It is beneficial from the suggested global-to-local search scheme, which mainly includes the asymmetric flip-based solution generation strategy, the SAD-based solution repair strategy, the perturbation based local search, and the endmember repetition-based solution selection strategy. Therefore, in this section, we empirically verify the effectiveness of these four strategies. Specifically, we compare the original GL-EA with GL-EA-noAF, GL-EA-noSR, EA-noLS, and GL-EA-noER. The GL-EA-noAF is the same as GL-EA, except that the suggested asymmetric flip-based solution generation strategy is replaced by the traditional crossover and mutation strategy in NSGA-II. The GL-EA-noSR is the same as our GL-EA, except that the SAD-based solution repair strategy is removed, which means that only the asymmetric flip operator is performed in the global search stage. Similarly, in GL-EA-noLS, the perturbation-based local search is deleted from GL-EA. For the fourth variant GL-EA-noER, the endmember repetition-based solution selection is replaced by the nondominated sorting and crowding distance sorting in NSGA-II.

The experimental results between GL-EA and its four variants are compared in terms of PFs and HV values, which are shown in Fig. 15 and Table VI. From Fig. 15, we can find that on different datasets, four GL-EA's variants have different performances. However, none of them could achieve better PFs on all three datasets. On the contrary, Fig. 15 reveals that PFs obtained by our GL-EA are always better than the four variants on all the sets. Especially on the dataset with the maximal spatial dimension (Urban), the superiority of GL-EA is more distinct. In addition, the statistics in Table VI also indicate

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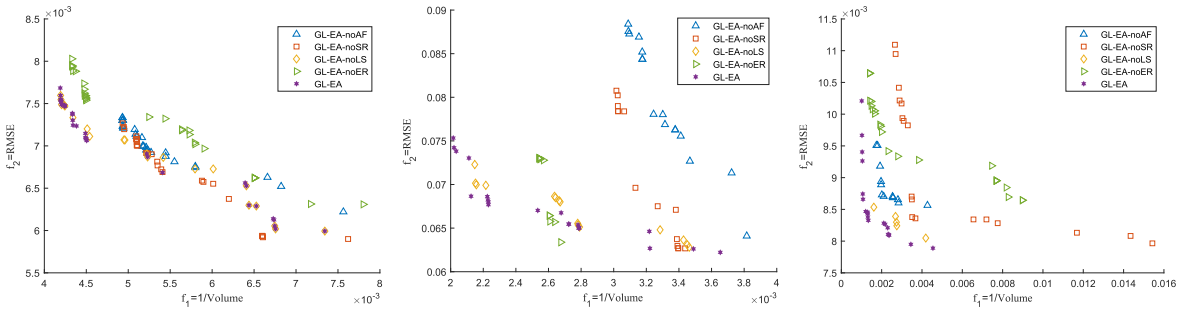

Fig. 15. PF results between GL-EA and its four variants on the three datasets. (a) Samson. (b) Jasper Ridge. (c) Urban.

TABLE VI
HV VALUES BETWEEN GL-EA AND ITS FOUR VARIANTS ON THE THREE DATASETS

Data sets	Samson	Jasper Ridge	Urban
GL-EA-noAF	0.2283	0.1859	0.5781
GL-EA-noSR	0.2448	0.2054	0.5792
GL-EA-noLS	0.2626	0.2245	0.6021
GL-EA-noER	0.2388	0.2143	0.5746
GL-EA	**0.2639**	**0.2330**	**0.6204**

that compared with its four variants, GL-EA has the highest HV values on all the datasets. All the comparison results above justify the effectiveness of four proposed strategies in GL-EA.

Finally, for further analyzing the performance of the proposed GL-EA, a series of additional experiments is also conducted from different aspects. These experiments mainly include three parts. 1) the running time comparison of different MOEA-based EE algorithms; 2) the abundance distribution maps of GL-EA on the experimental sets; and 3) the sensitivity analysis on parameter K in GL-EA. Due to space limitations, all of them are presented in the Supplementary Materials.

V. CONCLUSION

In this article, a GL-EA for hyperspectral EE was proposed. In GL-EA, to solve the sparse large-scale and constrained optimization problem of hyperspectral EE, the optimal solutions were approximated in a gradual way, where global search and local search were performed sequentially. Specifically, an asymmetric flip-based solution generation and an SAD-based solution repair strategies were developed in the global search phase. After that, a perturbation-based local search was designed. In addition, during the global search and local search phases, an endmember repetition-based solution selection strategy was also proposed to choose better solutions for the next iteration. The experimental results on different hyperspectral datasets have shown the superiority of the proposed GL-EA over the state-of-the-art EE methods.

The suggested global-to-local evolutionary method has shown its competitiveness in solving EE problems, and however, there still exist some interesting works to be further investigated. For example, the proposed GL-EA adopts population-based evolution, and during the evolution, each individual should be evaluated many times. Thus, its computational efficiency is not high, which is also a common problem in the existing EA-based algorithms. How to design an efficient EA for the EE problem is one of our future work. Moreover, in this work, we solve the sparse large-scale EE problem by designing the global-to-local search scheme. However, in the recent progress of evolutionary computation, there are several works that focus on other sparse large-scale applications (such as feature selection [60] and instance selection [61]). Thus, how to extend these works to solve the EE problem is also one of our future works.

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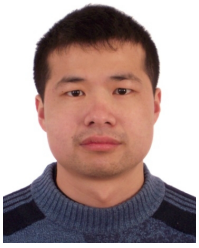

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